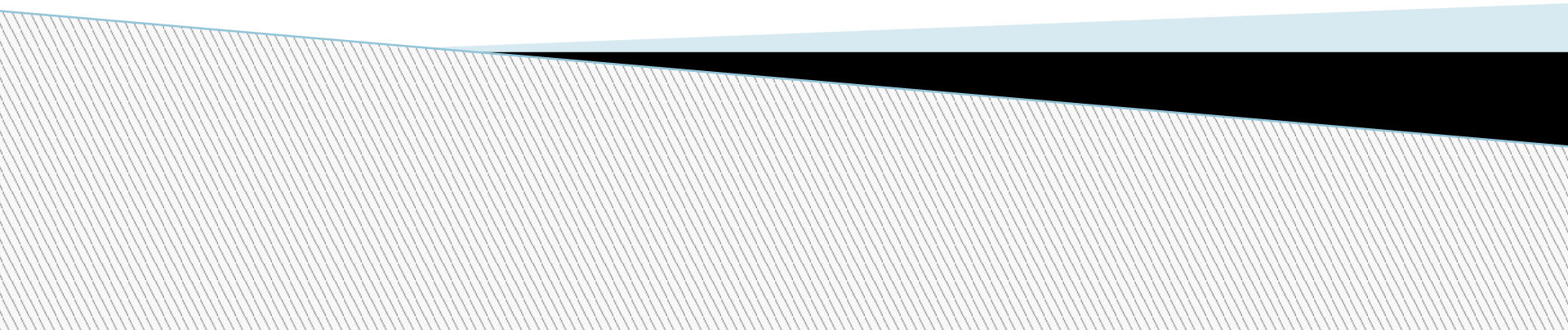
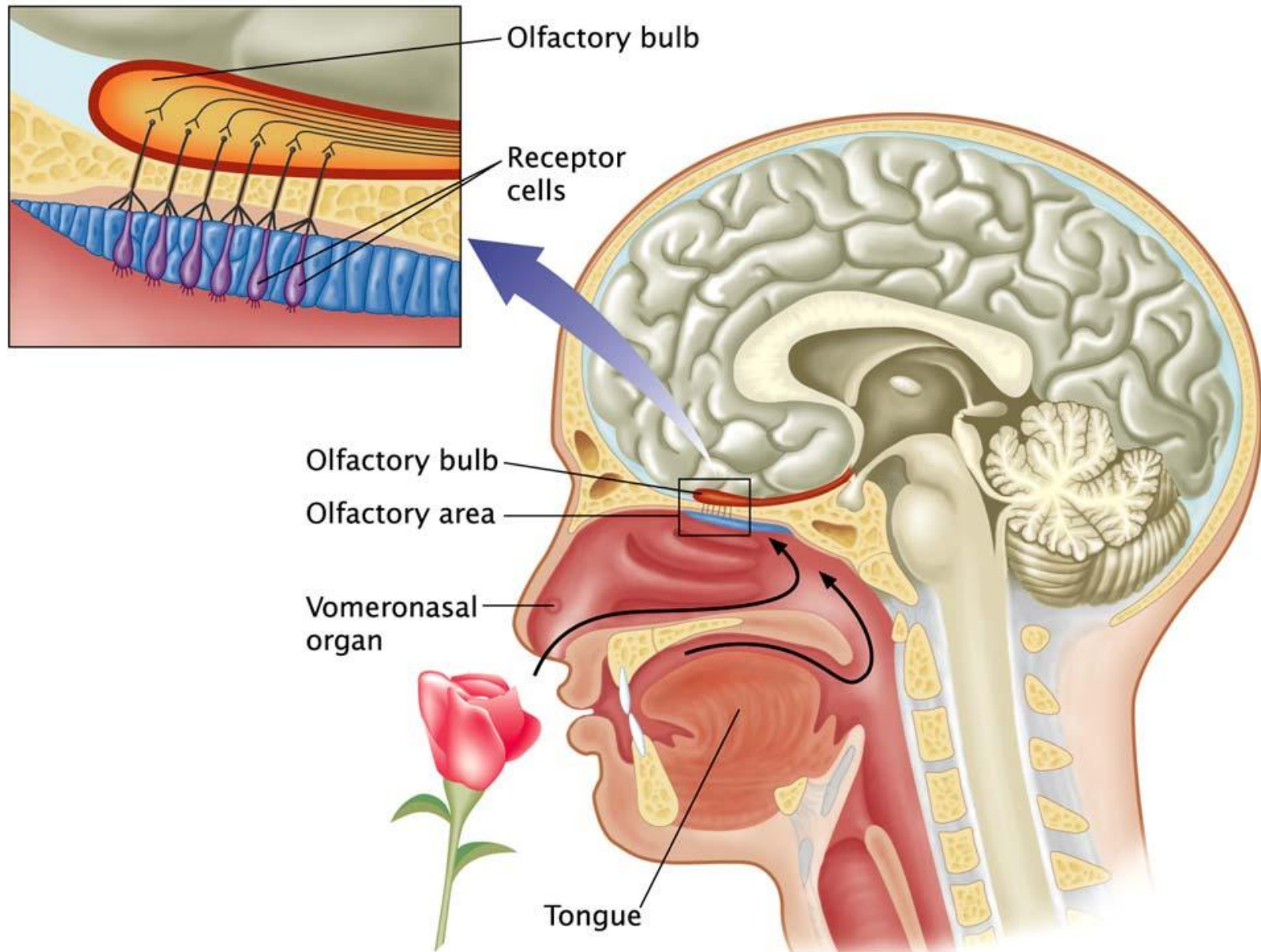


SENSE OF OLFACTION

DR SHAHNEELA SIRAJ
DIKIOHS,DUHS.

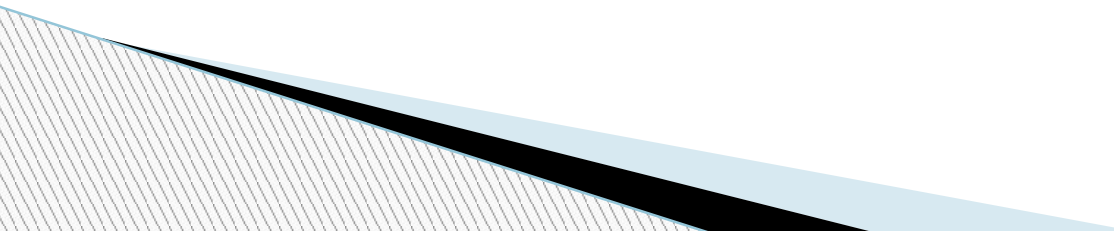


The Olfactory System

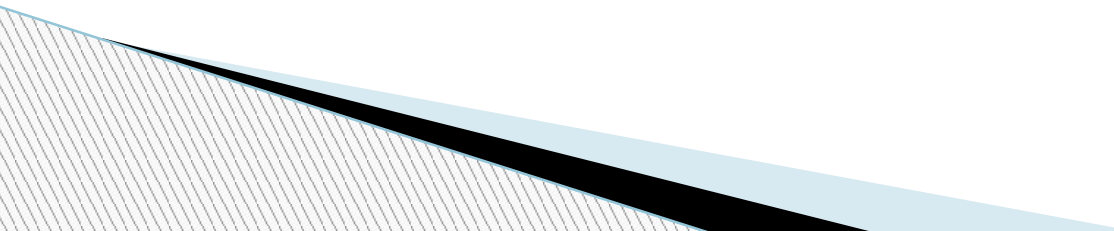


LEARNING OBJECTIVES

At the end of lecture, students should be able to

- Describe Olfactory receptors
 - Explain Olfactory pathway
 - Discuss Olfactory Transduction
 - Enumerate the abnormalities of olfaction.
- 

Important Terms

- ▣ **Sense:** A system that translates information from outside the nervous system into neural activity
 - ▣ **Sensations:** Messages from the senses
 - ▣ **Perception:** The process of making sensations into meaningful experiences (conscious awareness of a sensation)
 - ▣ **A sensory receptor** :-- specialized cell that monitors conditions in the body or the external environment
 - ▣ Stimulation of the receptor □ alters the production of action potentials in a sensory neuron
- 

The Chemical Senses

- ▣ **Olfaction** detects airborne chemicals
Our sense of smell
- ▣ **Gustation** detects chemicals in solution that come into contact with receptors inside the mouth

The Special Senses—Smell

▣ **Olfactory Organs**

- Olfactory epithelium

 - Olfactory receptor cells

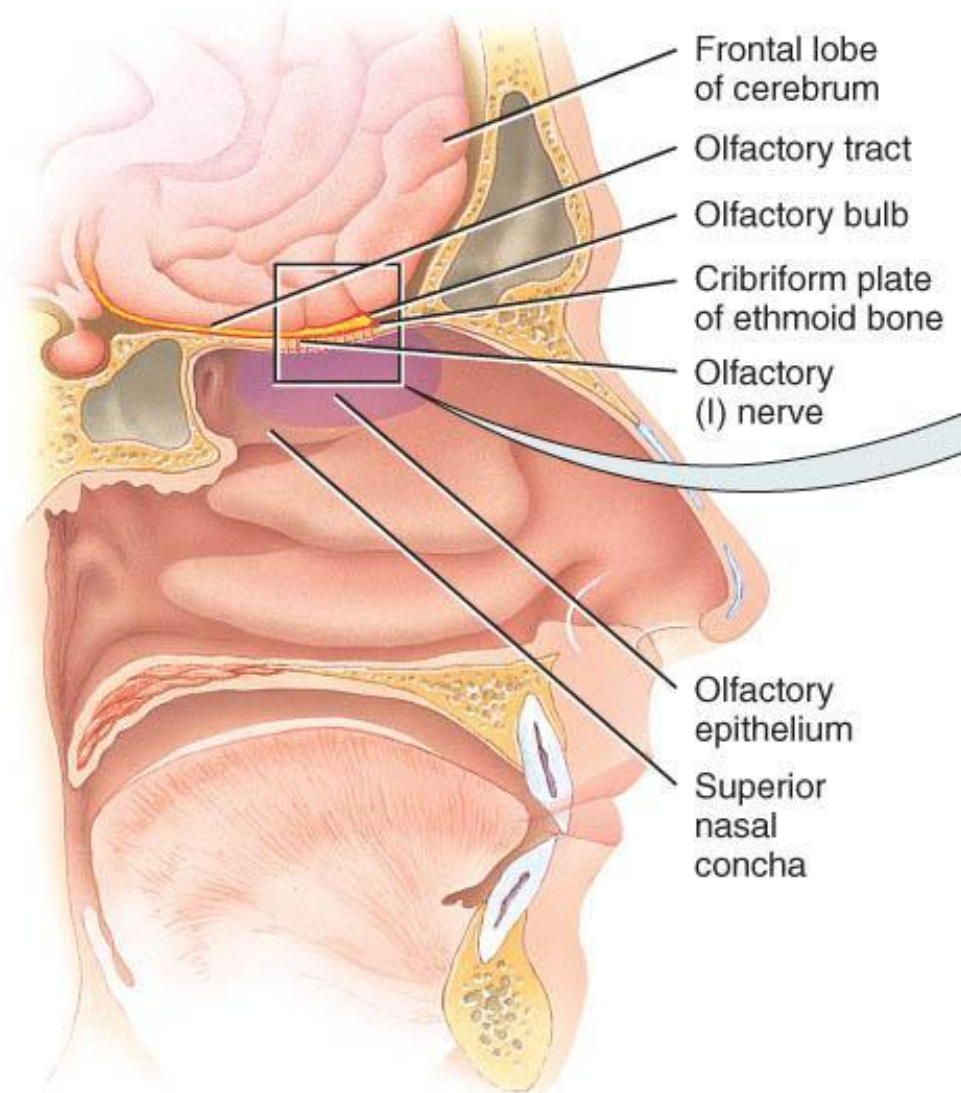
 - Neurons sensitive to *odorants*

 - Supporting cells

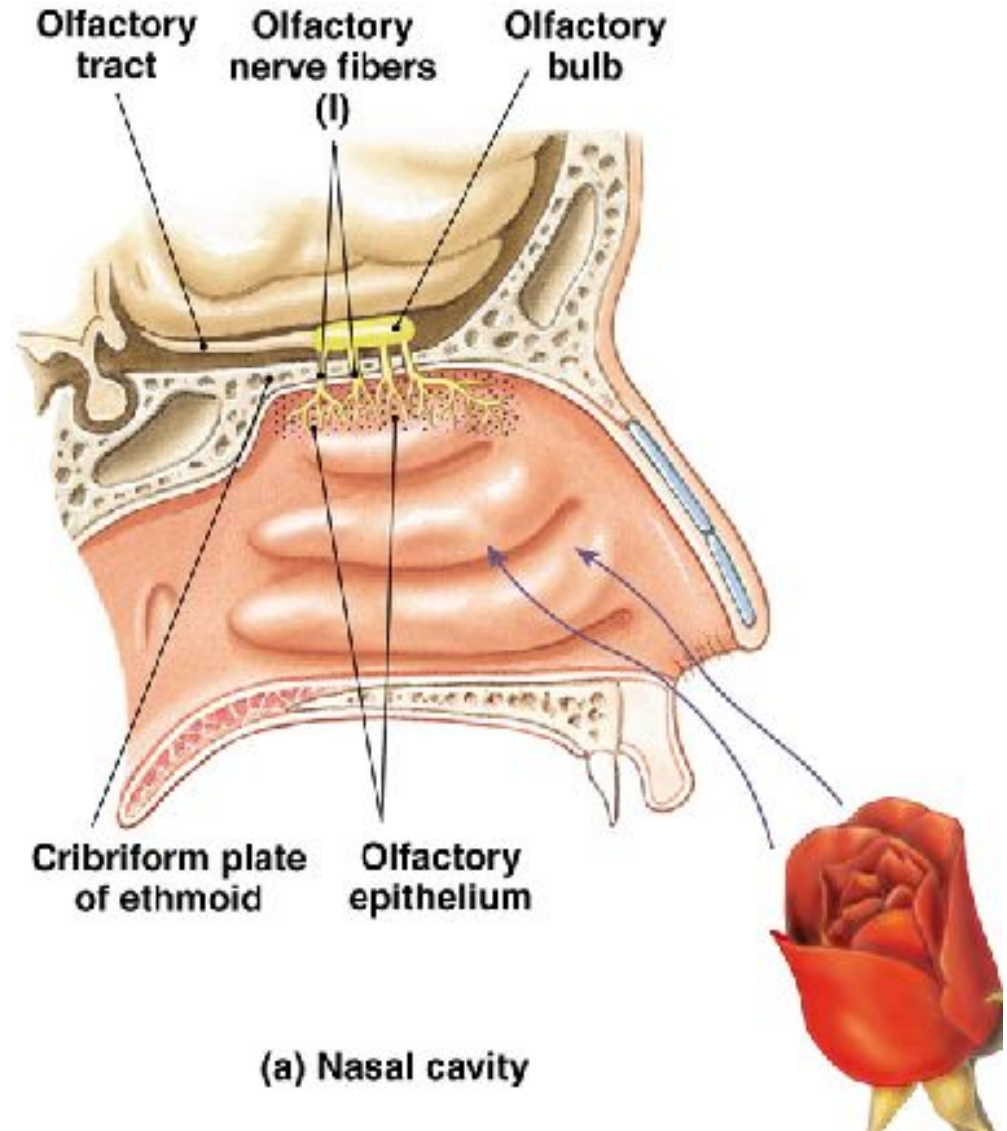
 - Basal (stem) cells

Olfactory Epithelium

- 3 square centimetre of membrane holding 10-100 million receptor cells
- Covers superior nasal cavity and cribriform plate
- 3 types of receptor cells

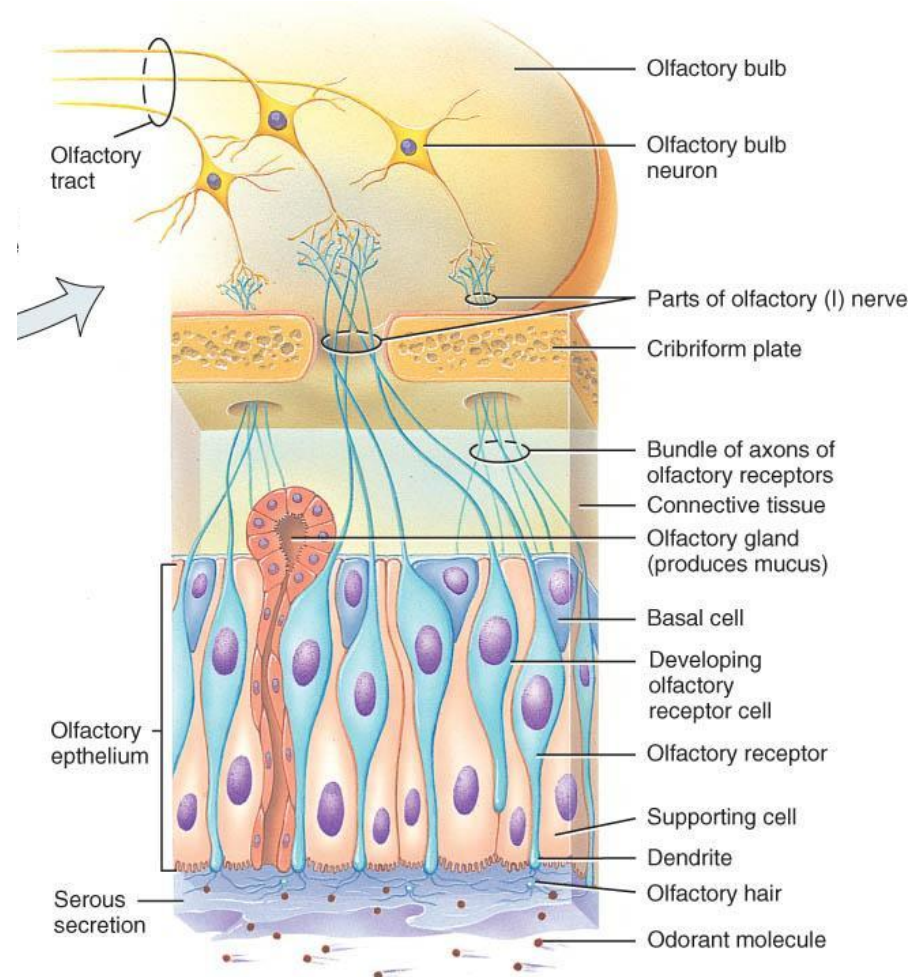


□ The Olfactory Organs

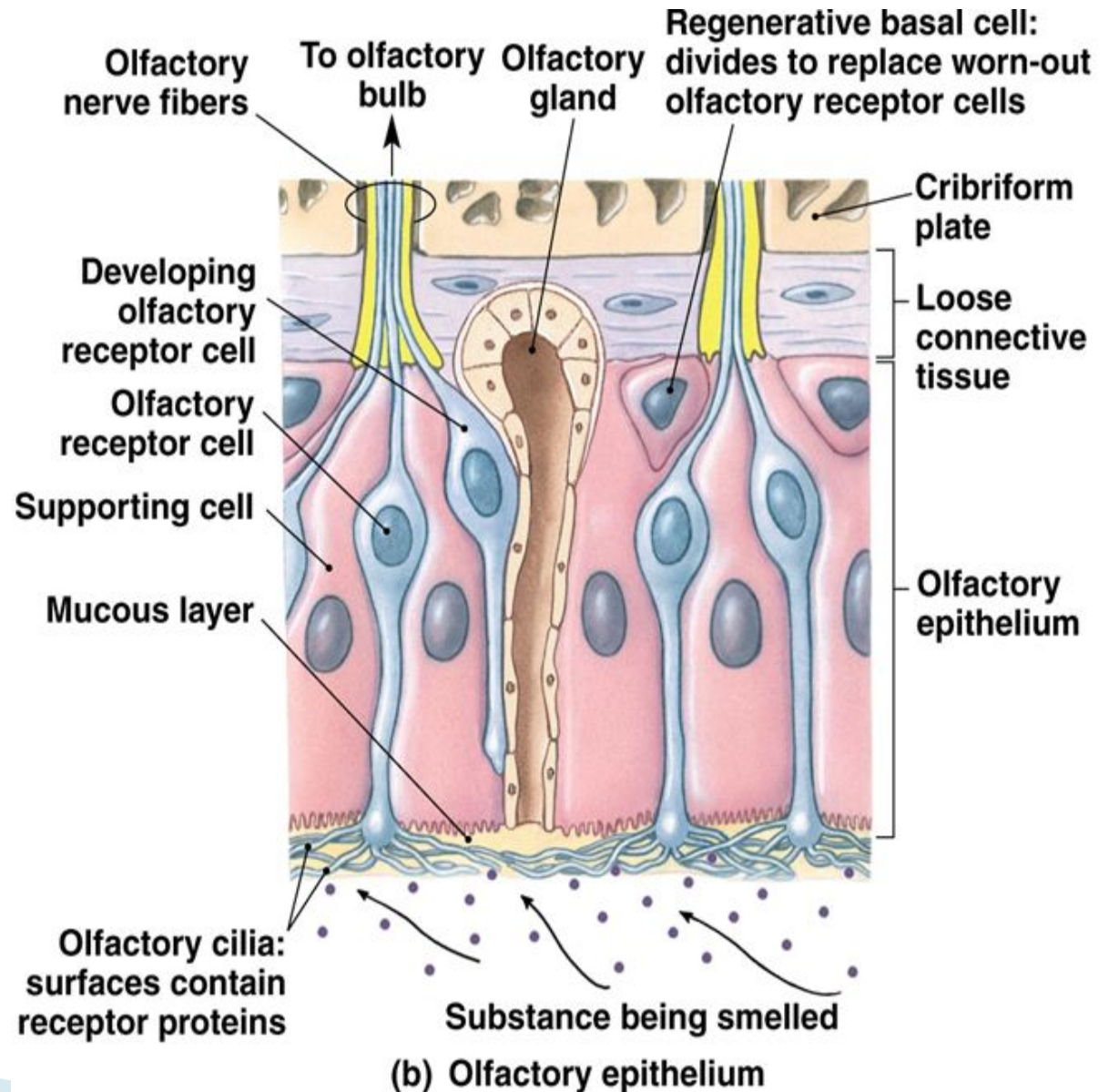


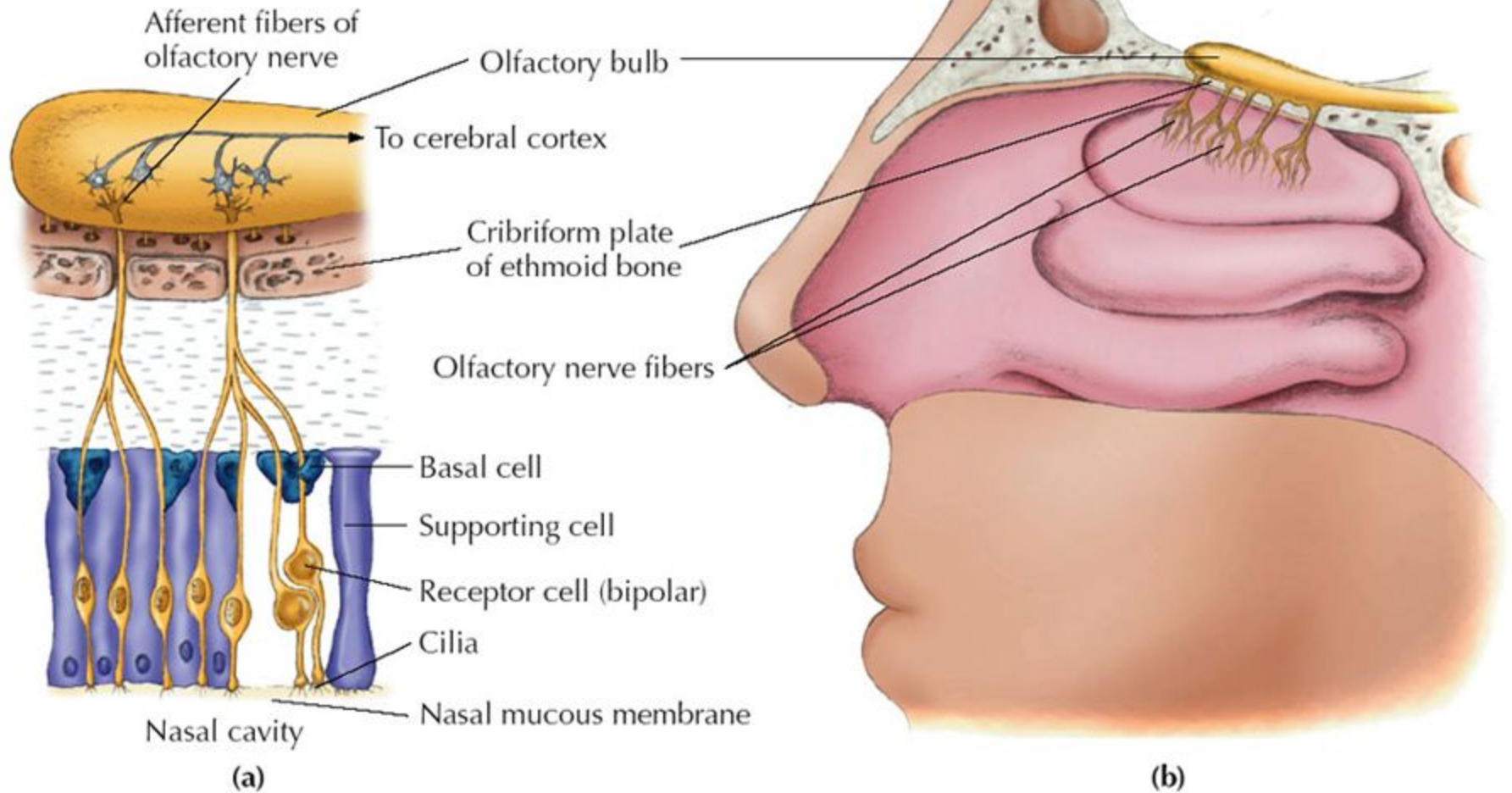
Cells of the Olfactory Membrane

- ❑ **Olfactory receptors**
bipolar neurons with cilia or olfactory hairs
- ❑ **Supporting cells**
columnar epithelium
- ❑ **Basal cells = stem cells**
replace receptors monthly
- ❑ **Olfactory glands**
produce mucus



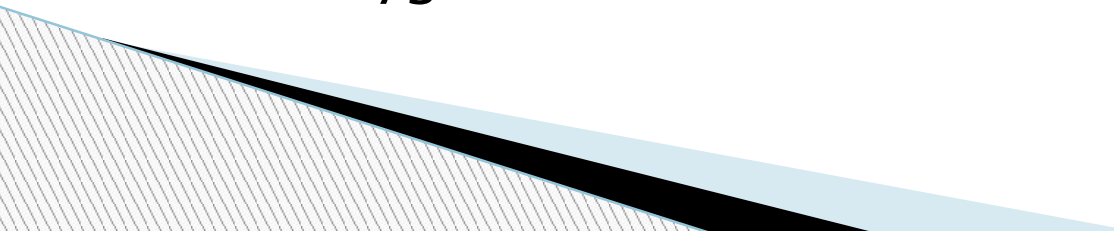
□ The Olfactory Organs

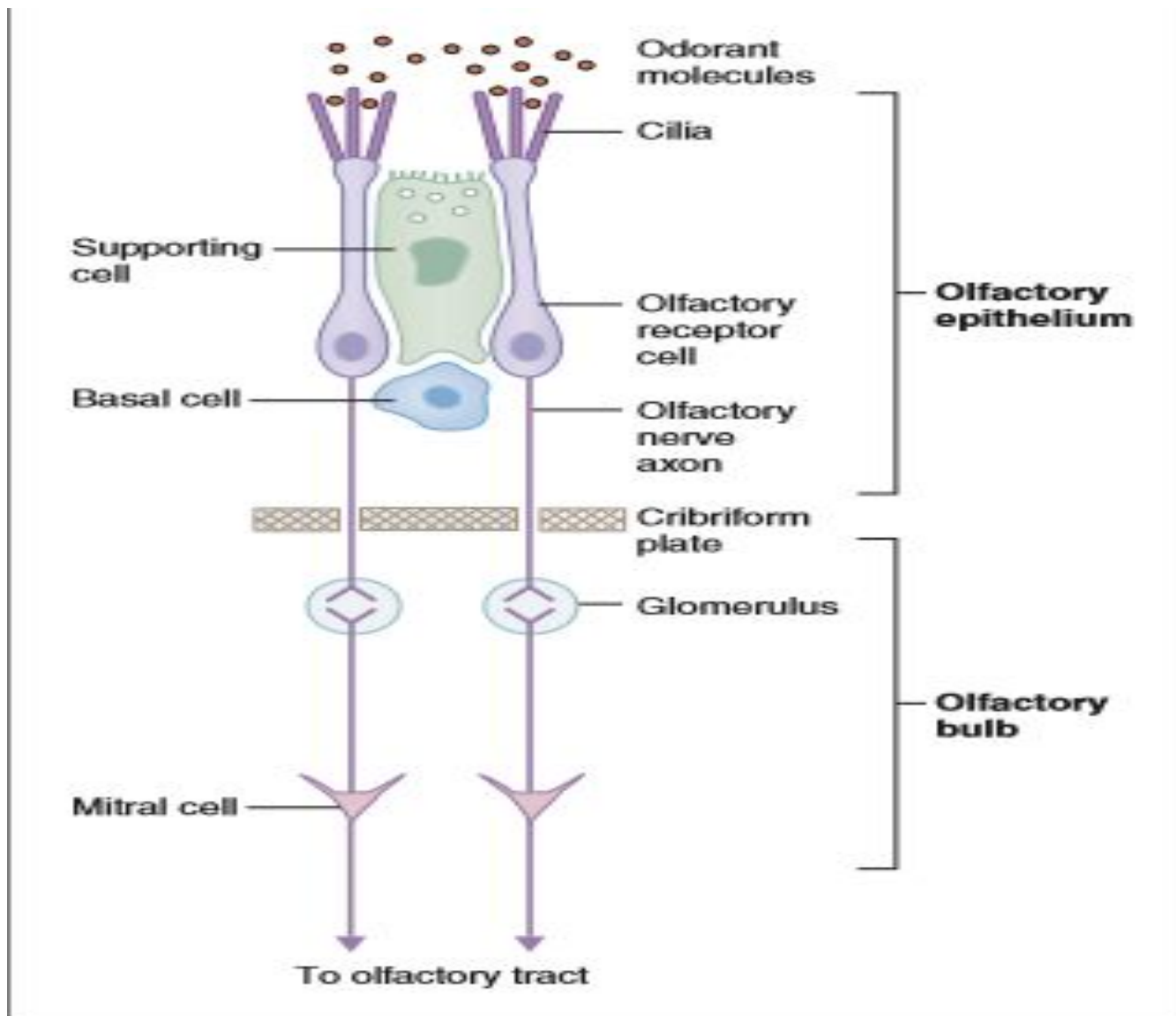




(a) Olfactory nerve fibers respond to gaseous molecules. Receptor cells are shown in cross section to the left. (b) Olfactory receptors are located in the upper nasal cavity. (c) On the right, an extreme close-up of an olfactory receptor shows fibers that sense gaseous molecules of various shapes.

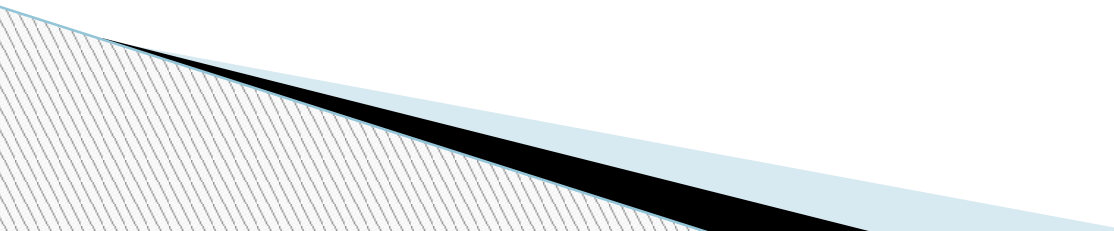
OLFACTORY PATHWAY

- ❑ Axons of bipolar olfactory receptors pierce the cribriform plate of ethmoid bone and reach the olfactory bulb. Here, the axons synapse with dendrites of mitral cells. Different groups of these synapses form globular structures called **olfactory glomeruli**.
 - ❑ Axons of mitral cells leave the olfactory bulb and form olfactory tract. Olfactory tract runs backward and ends in olfactory cortex, through the intermediate and lateral olfactory stria.
 - ❑ Olfactory cortex includes the structures, which form a part of limbic system. These structures are anterior olfactory nucleus, prepyriform cortex, olfactory tubercle and amygdala.
- 



Olfactory pathways, showing the olfactory epithelium and olfactory bulb

OLFACTORY PATHWAY

- One path leads from the olfactory bulbs via the olfactory tracts to the olfactory cortex where smells are consciously interpreted and identified
 - Another path leads from the olfactory bulbs via the olfactory tracts to the thalamus and limbic system where smells elicit emotional responses
 - Smells can also trigger sympathetic nervous system activation or stimulate digestive processes
- 

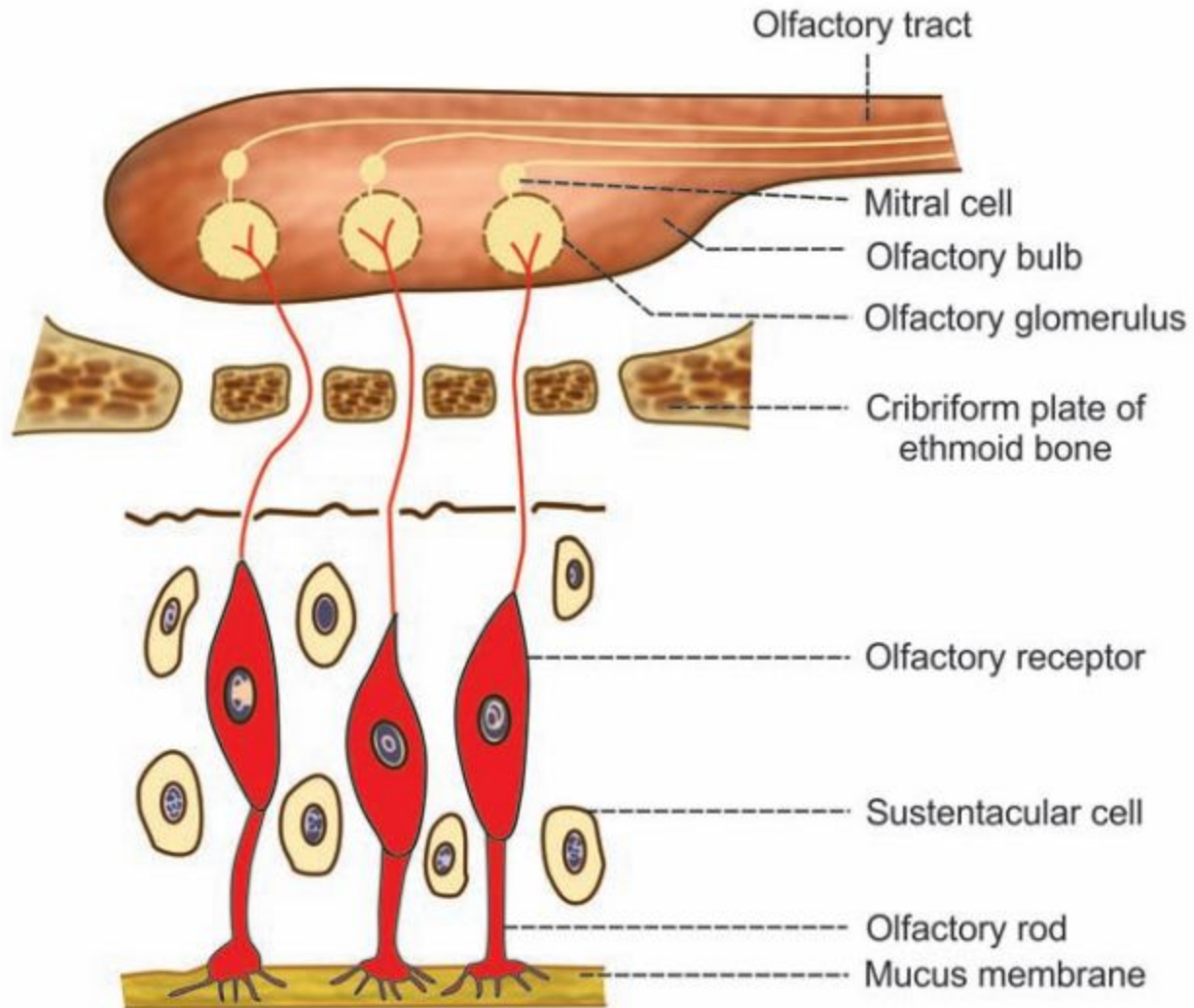
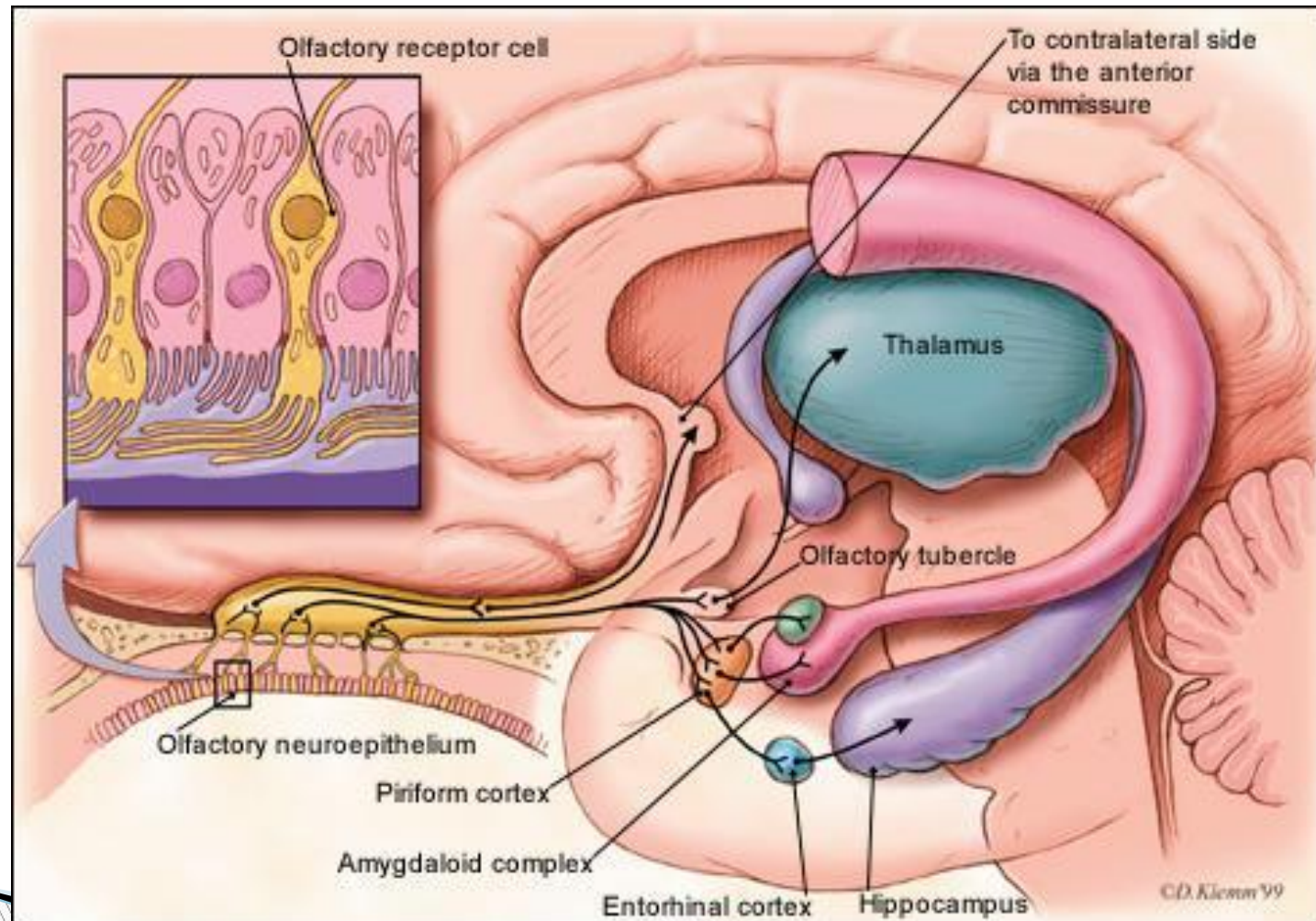


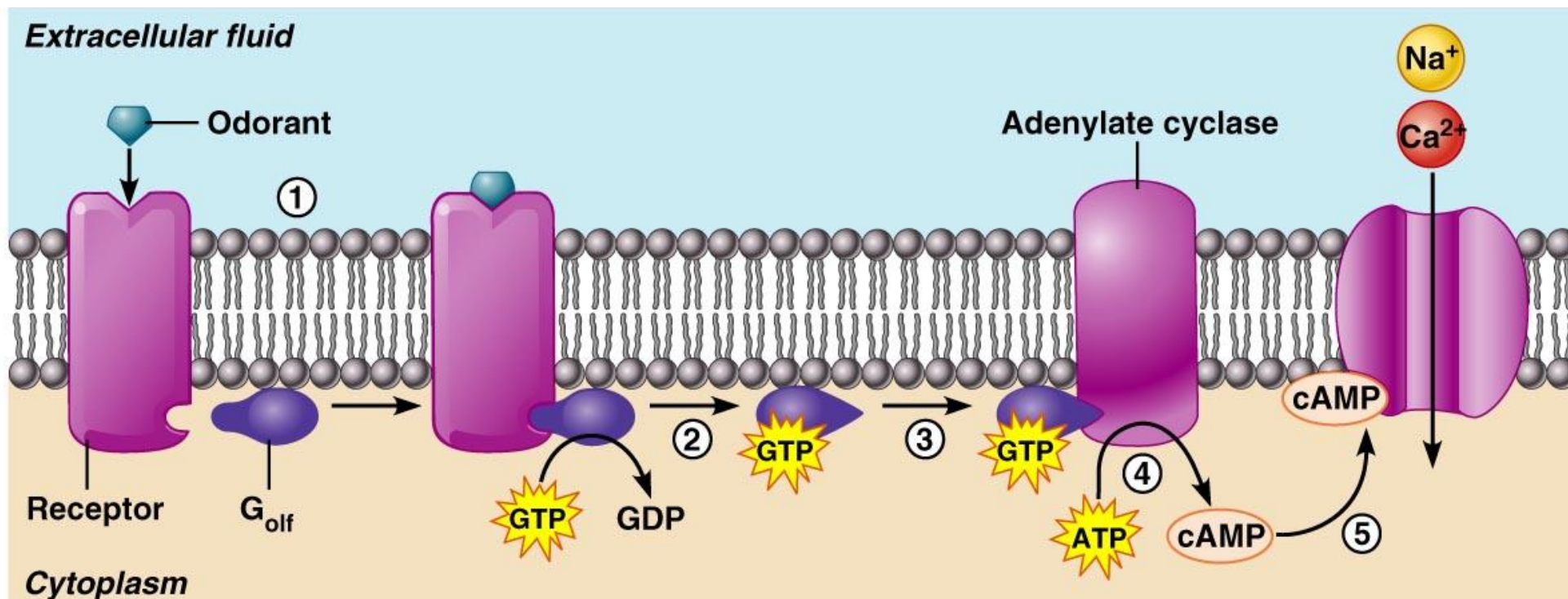
FIGURE 177.1: Olfactory mucus membrane and pathway for olfactory sensation

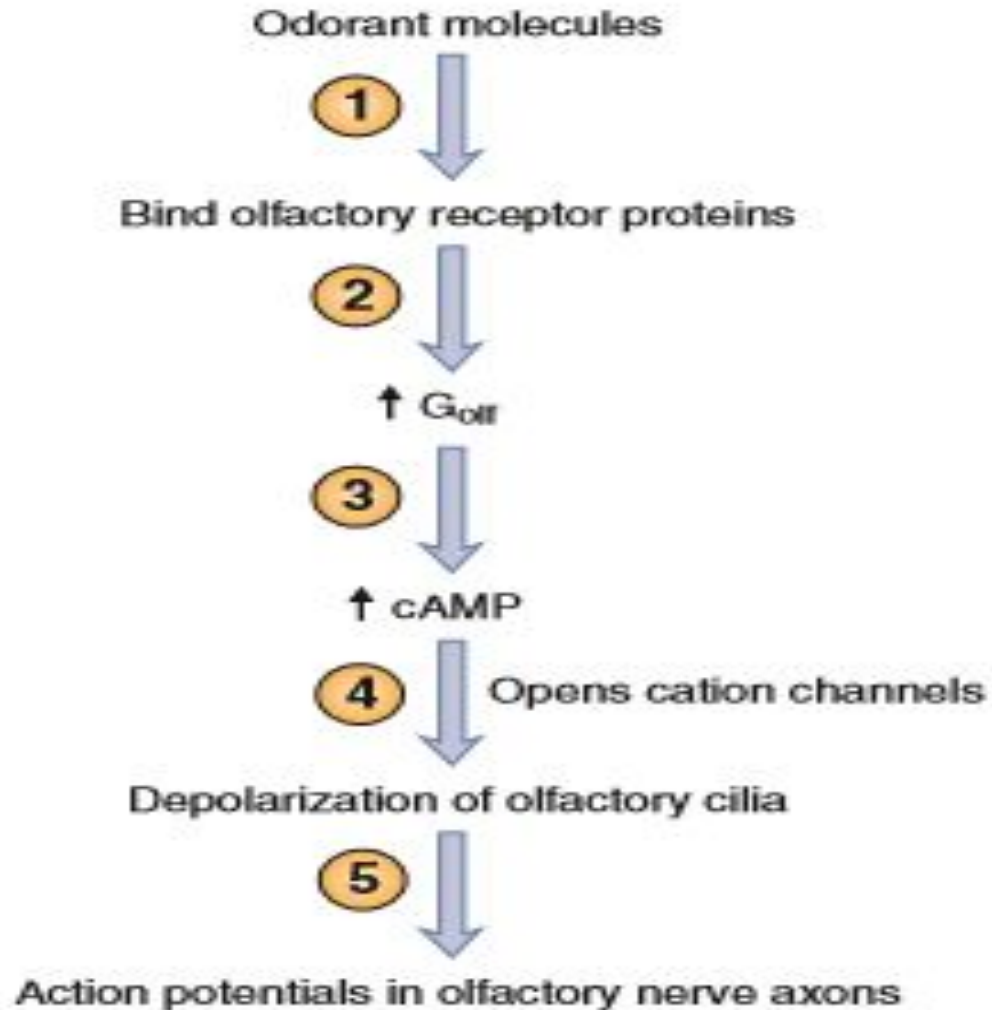
Cortical Regions Associated with Olfactory Information



Transduction of Smell

- Binding of an odorant molecule to a specific receptor activates a G-protein and then a second messenger (cAMP)
- cAMP causes gated Na^+ and Ca^{2+} channels to open, leading to depolarization





Steps in olfactory transduction

OLFACTORY TRANSDUCTION

- ❑ Olfactory transduction is the process by which olfactory receptor converts chemical energy into action potentials in olfactory nerve fiber.
- ❑ The odoriferous substance stimulates the olfactory receptors, only if it dissolves in mucus, covering the olfactory mucus membrane.
- ❑ Molecules of dissolved substance, bind with receptor proteins in the cilia and form substance-receptor complex. Substance_x0002_receptor complex activates adenyl cyclase that causes the formation of cyclic AMP.
- ❑ Cyclic AMP in turn, causes opening of sodium channels, leading to influx of sodium and generation of receptor potential.
- ❑ Receptor potential causes generation of action potential in the axon of bipolar neuron.

Stimulation of Olfactory cells

- ▣ **Stimulation of Olfactory cells**
- ▣ G-protein is stimulated -triggers activation of Adenyl cyclase (enzyme speeds up the conversion of ATP to cAMP – cAMP then binds to action channels in membrane of cilia- this causes channels to open and Ca ions to enter cilia – influx of Ca ions activates Cl channels to open and Cl leaves. Membrane becomes depolarized and AP is created. The action potential travels down the axon of olfactory receptor cell eventually meets with the other axons and forms the olfactory nerve (CN I

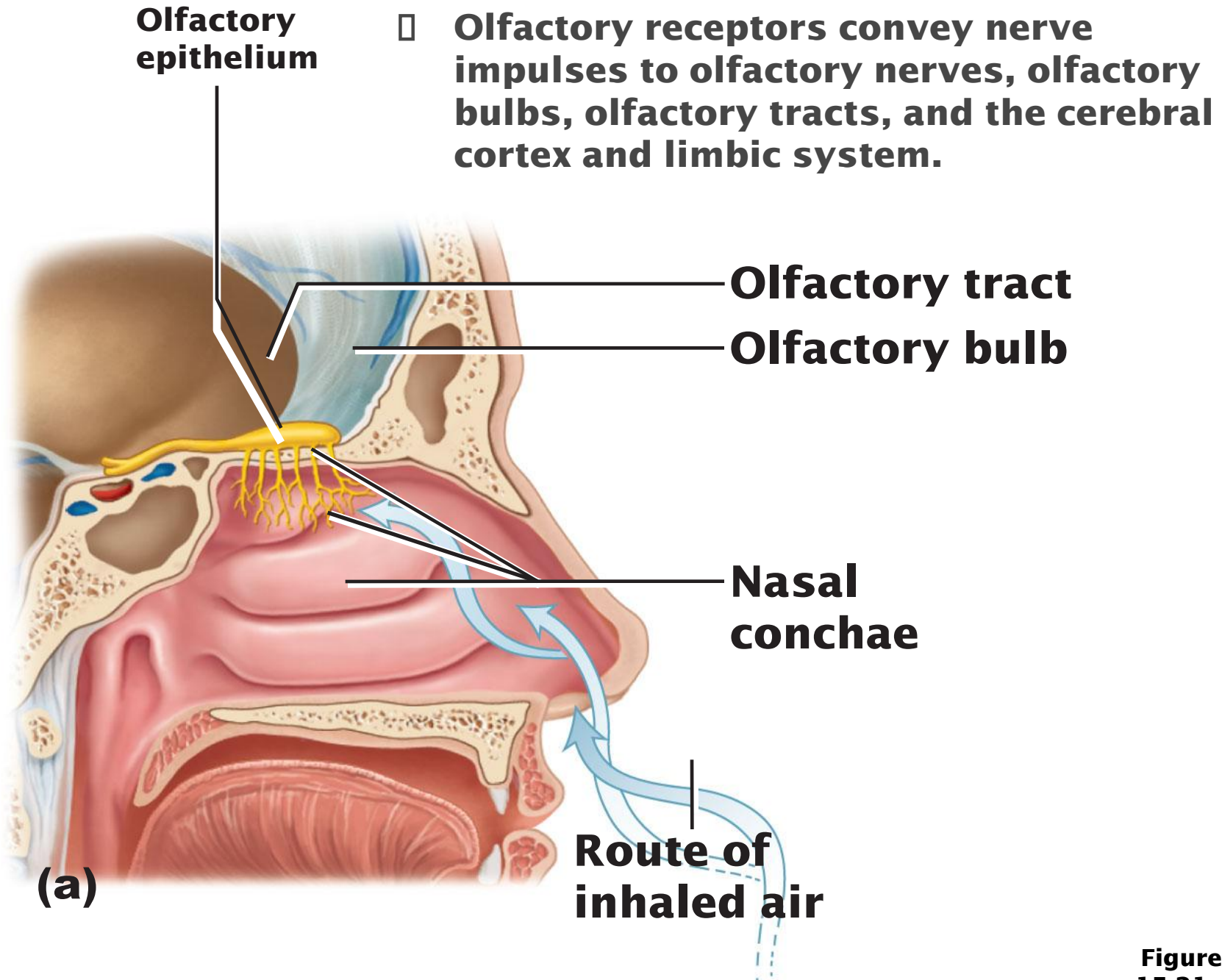
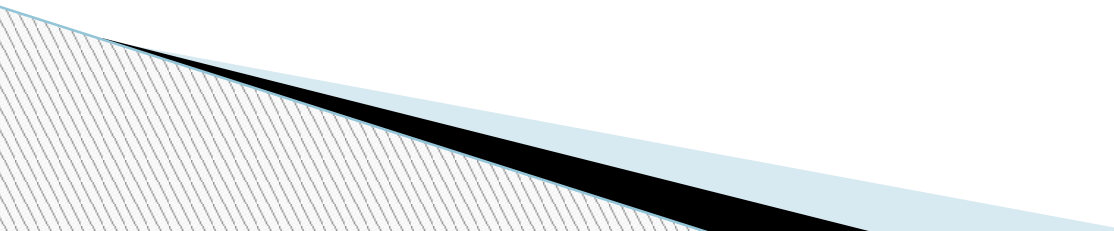


Figure
15.21

CLASSIFICATION OF ODOUR

- Odor is classified into various types. Substances producing different types of odor are:
 1. Aromatic or resinous odor: Camphor, lavender, clove and bitter almonds
 2. Ambrosial odor: Musk
 3. Burning odor: Burning feathers, tobacco, roasted coffee and meat
 4. Ethereal odor: Fruits, ethers and beeswax
 5. Fragrant or balsamic odor: Flowers and perfumes
 - 6. Garlic odor: Garlic, onion and sulfur
 - 7. Nauseating odor: Decayed vegetables and feces
 - 8. Repulsive odor: Bed bug.

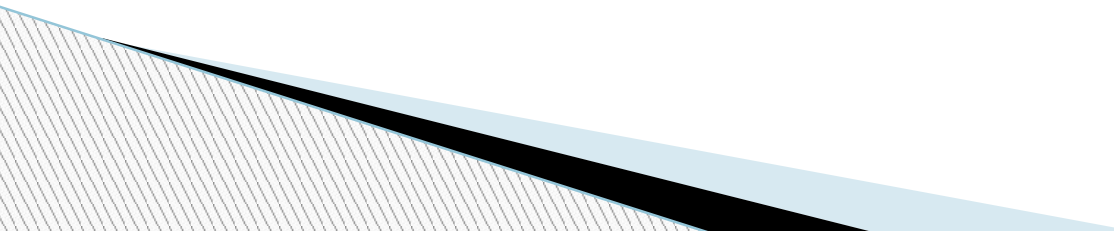
IMPORTANCE OF SMELL

- ❑ Important for enjoyment and selection of food.
 - ❑ Flavors are combination of taste and smell (smell contribution is about 80%)
 - ❑ Gives warning of harmful substances or places.
- 

ODOUR THRESHOLDS AND ADAPTATION

- ❑ Chemical METHYL MERCAPTAN, which smells like rotten cabbage and can be detected in concentration as low as 1/25 billionth of a milligram per millilitre of air.
- ❑ It is added to natural gas to detect gas leaks.
- ❑ **ADAPTATION**

Olfactory receptors are phasic receptors and adapt very rapidly. Within one second, the adaptation occurs upto 50%.



OLFACTORY DISORDERS

Anosmia

Anosmia refers to total loss of sensation of smell, i.e. inability to recognize or detect any odor. It may be temporary or permanent. **Temporary anosmia** is due to obstruction of nose, which occurs during common cold, nasal sinus and allergic conditions. **Permanent anosmia** occurs during lesion in olfactory tract, meningitis and degenerative conditions such as Parkinson disease and Alzheimer disease.

Hyposmia

Hyposmia is the reduced ability to recognize and to detect any odor.

Hyperosmia

Hyperosmia is the increased or exaggerated olfactory sensation.

